Year 12 Maths - Trigonometric functions and equations

Similar questions - Cubics and inequalities in trig

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 $f(x) = 4x^3 - 4x^2 - 5x + 3$.

(a) Show that $(x+1)$ is a factor of $f(x)$ and factorise $f(x)$ completely. **[4]**

(b) Hence solve $4\sin^3\theta - 4\sin^2\theta - 5\sin\theta + 3 = 0$ for $0^\circ \leqslant \theta < 360^\circ$. **[4]**

(c) Explain why one of the three roots of $f(x) = 0$ does not lead to any value of θ. **[1]**

2

(a) Show that the inequality $2\sin^2\theta > 3\cos\theta$ can be written as $2\cos^2\theta + 3\cos\theta - 2 < 0$. **[2]**

(b) Hence solve $2\sin^2\theta > 3\cos\theta$ for $0^\circ \leqslant \theta \leqslant 360^\circ$. **[4]**

(c) Without further working, state whether $\theta = 180^\circ$ satisfies the inequality, and check your answer by substituting. **[1]**

Question 1

(a) $f(-1) = -4-4+5+3 = 0$, so $(x+1)$ is a factor. $f(x) = (x+1)(4x^2-8x+3) = (x+1)(2x-1)(2x-3)$

(b) $\sin\theta = -1$, $\frac{1}{2}$ or $\frac{3}{2}$. $\sin\theta = -1$ gives $\theta = 270°$; $\sin\theta = \frac{1}{2}$ gives $\theta = 30°, 150°$. So $\theta = 30°, 150°, 270°$

(c) $\sin\theta = \frac{3}{2} > 1$ is impossible

Question 2

(a) $2(1-\cos^2\theta) > 3\cos\theta$, so $0 > 2\cos^2\theta + 3\cos\theta - 2$

(b) $(2\cos\theta - 1)(\cos\theta + 2) < 0$. As $\cos\theta + 2 > 0$ always, need $2\cos\theta - 1 < 0$, so $\cos\theta < \frac{1}{2}$. This holds for $60° < \theta < 300°$

(c) Yes: $2\sin^2 180° = 0 > 3\cos 180° = -3$